

Instrumentality were significantly correlated with P(I), whereas the variables of T and S were not. There is a need to test the model further in a real world situation and to relate it to the actual degree of adoption of an innovative strategy.

The semantic differential instrument used in this field study is both simple and versatile to use. It provides a ready handle on the current status of T, S, C, R, P_N, and P_S in an organizational setting. A manager could use this instrument to obtain feedback on how his group or department stands in relation to a proposed innovation. Further, Model 1 of this study provides clear conceptual guidance as to the interaction of the above variables and their joint effect on the probability of adopting an innovative strategy. With knowledge of the current status of the above key variables, the manager is able to observe which variables are most unfavorable to the adoption of an innovative strategy and to be able to consider changes accordingly.

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BOUNDARY-SPANNING ROLES IN A RESEARCH AND DEVELOPMENT ORGANIZATION: AN EMPIRICAL INVESTIGATION¹

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The importance of boundary-spanning roles that serve to link organizations has been confirmed by research in a number of diverse organizations (1, 2, 12, 16, 19, 20). These roles are vital to the effective monitoring of the environment and to the transfer of technology and information across boundaries.

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Considerable prior research (6, 9, 10, 13, 18), however, has found boundary-spanning roles to have negative effects for their incumbents. Since boundary spanners have *role senders* located in external organizations as well as in their own organization, conflicting and misunderstood expectations for role performance often are sent to the boundary spanner. These conditions have been found to result in strong and conflicting role pressures and tensions for incumbents of boundary-spanning roles.

Marginality also has been found to be higher for those in boundary-spanning roles (7, 14, 23). Marginality is a self-orientation concerning two or more groups of others: a person with high marginality would stand between two or more groups with different value systems; a person with low marginality would be associated with a single group. Those high on marginality have been found to be better able to deal objectively with two or more groups and to be more satisfied and effective in boundary positions than are nonmarginal individuals. Organ (8) has also hypothesized that effective boundary spanners should have relatively high economic and political value orientations, and relatively low aesthetic and religious value orientations. The hypothesized value orientations are due to the pragmatism and "hypocrisy" required of boundary spanners.

In the present study the Kahn et al. (6) theory of role dynamics was used to generate the research hypotheses. It was hypothesized that boundary-spanning activity (BSA) would be positively related to levels of role conflict and ambiguity, and negatively related to levels of job satisfaction. It was also hypothesized that BSA would be positively related to marginality and to economic and political value orientations, and negatively related to aesthetic and religious value orientations.

Method

The study was conducted in an applied science department of a large government research and development organization. This department was selected because of its external interfaces with more than 20 universities, private contractors, and other governmental agencies, as well as its many positions with primarily internal relationships. The sample of 51 professional employees (a 58 percent response rate) was similar to its population in age, work experience, and education. All subjects had at least a bachelor's degree, and most had a graduate degree. Anonymous questionnaires were returned by mail directly to the researchers, and interviews were conducted with many of the boundary spanners to corroborate the questionnaire data.

To measure the level of BSA, a four item scale was constructed based on items successfully used in prior research (1, 2, 4). Each of the following items had a five point scale of 0, 1-2, 3-4, 5-6, and more than 6:

- (a) Indicate the *total* number of magazines, journals, and newspapers which you regularly read.
- (b) During the last month, what was the frequency with which you recommended specific information sources

(e.g., journal article, knowledgeable colleague, report, magazine, etc.) to a colleague? (c) During the last month, indicate the frequency with which *you have sought* information/advice from members of other organizations outside (name of organization). (d) During the last month, indicate the frequency with which members of organizations outside (name of organization) have sought information/advice *from you*.

The BSA score was the sum of the four items. To validate the BSA scale, top management of the department identified background characteristics of personnel thought to be high on BSA. All those so identified scored in the top third of the BSA scale distribution; moreover, a *t*-test between the BSA scale means of those identified as boundary spanners and others was highly significant ($t = 4.07, p < .001$). The reliability (Coefficient Alpha) of the BSA scale was .73; the average score was 11.7, and the standard deviation was 4.0.

Role conflict and ambiguity were measured by the two Rizzo et al. (15) scales. The Job Description Index (17) was used to measure five dimensions of job satisfaction: work itself, co-workers, supervision, pay, and opportunities for promotion. The marginality measure was that used by Ziller et al. (23), and six personality related values were measured by the Study of Values test (3).

Results

Tables 1 and 2 report the results of a correlational analysis. Contrary to the hypotheses, BSA was negatively related to role ambiguity and positively related to job satisfaction with co-workers, pay, and opportunities for promotion. In addition, BSA was not related to marginality or any of the personality-related values; thus, these hypotheses were not supported either.

Partial correlations holding marginality constant also were computed to eliminate effects introduced by both high and low marginality types being attracted to boundary-spanning roles. These data, reported in Tables 3 and 4, do not provide support for any of the research hypotheses.

TABLE 1
Correlations Between Boundary-Spanning Activity and Role and Satisfaction Variables

	Role Conflict	Role Ambi- guity	Dimensions of Job Satisfaction				
			Work	Co- Workers	Super- vision	Pay	Promo- tions
Boundary-Spanning Activity	.08	-.31*	.24	.41**	-.11	.28*	.43**

* $p < .05$

** $p < .01$

TABLE 2
Correlations Between Boundary-Spanning Activity and
Marginality and Value Variables

	<i>Marginality</i>	<i>Personality-Related Values</i>					
		<i>Theoretical</i>	<i>Economic</i>	<i>Aesthetic</i>	<i>Social</i>	<i>Political</i>	<i>Religious</i>
Boundary-Spanning Activity	.00	.23	.09	.09	-.01	-.20	-.13

TABLE 3
Partial Correlations Between Boundary-Spanning Activity and
Role and Satisfaction Variables (Marginality Held Constant)

	<i>Role Conflict</i>	<i>Role Ambiguity</i>	<i>Dimensions of Job Satisfaction</i>				
			<i>Work</i>	<i>Co-Workers</i>	<i>Supervision</i>	<i>Pay</i>	<i>Promotions</i>
Boundary-Spanning Activity	.25	.20	.05	.65***	-.36**	.36**	.38**

* $p < .05$

** $p < .01$

*** $p < .001$

TABLE 4
Partial Correlations Between Boundary-Spanning Activity and
Value Variables (Marginality Held Constant)

	<i>Personality-Related Values</i>					
	<i>Theoretical</i>	<i>Economic</i>	<i>Aesthetic</i>	<i>Social</i>	<i>Political</i>	<i>Religious</i>
Boundary-Spanning Activity	.34*	.15	.06	.23	-.28*	-.32*

* $p < .05$

Discussion and Implications

When compared with prior research (6, 9, 10, 13, 18, 23), the present study indicates that boundary-spanning roles can differ substantially depending on the interorganizational relations and context. Where the organizations which are spanned have goals that are in conflict, the boundary spanner can experience strong role pressures and tensions due to the conflicting expectations for performance (e.g., a salesman or purchasing agent). In the present study, however, the interorganizational goals are relatively compatible. The research and development organization needs the tech-

nology and information supplied by the universities and contractors to solve its ambiguous problems. The universities and contractors, moreover, want to publicize their research findings and applications. Boundary-spanning roles here really serve to transfer technology and information that are used to solve problems and reduce ambiguities. And, as found by Pettigrew (11), boundary spanners are able to use their access to information as a power resource to obtain better pay, opportunities for promotion, and favorable relations with co-workers.

With low degrees of role pressures and stress, it is understandable that high marginality types or those with economic and political values are not necessarily attracted to the boundary-spanning roles in the present study. When marginality is held constant, the partial correlations between BSA and role conflict and ambiguity are in the predicted direction. It thus appears that both the low degrees of role pressures and stress, and the attraction of both high and low marginality types to these boundary-spanning roles can explain the differences in results between this study and prior research. The partial correlations also show BSA positively related to a theoretical value orientation, and negatively related to political and religious values. However, caution is urged in interpreting significance levels for the ipsatively scored Study of Values test (5). In effect, with marginality held constant, these boundary-spanning roles appear to attract those with a theoretical value orientation who seek the discovery of "truth," rather than those with a political value who are oriented toward power. These orientations are consistent with the typical scientist in a research and development organization.

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STATISTICAL PROBLEMS OF CONTINGENCY MODELS IN ORGANIZATIONAL BEHAVIOR¹

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Interest in the conditions under which the relationship between supervisor and subordinate results in effective work behavior has stimulated the development of a number of leadership theories. Although a number of the theories have achieved some success in accounting for these behaviors, relatively few have dealt with the coordinate question concerning the degree to which theoretical findings lend themselves to practical application by management. In this paper some of the methodological and statistical problems of contingency theories of leadership are examined from both of these perspectives.

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